

16-Week Live Interactive Program

Transform from a beginner to an **industry-ready** Gen AI Developer.

Master the end-to-end lifecycle of designing, building, and deploying advanced Generative AI solutions. Learn from industry experts and build production-grade applications.

★ 4.8 Rating

👤 1:1 Mentorship

🎧 24/7 AI Support

agent_loop.py

```
> INITIALIZING CREW AI...
> LOADING LLM (Ollama - Llama 3)...
> ESTABLISHING RAG PIPELINE...
> Connecting Vector DB... [OK]
> Deploying to Docker... [OK]

> SYSTEM READY.
```



The Complete Learning Lifecycle

We don't just teach theory. We take you from Python basics to deploying fully autonomous, cloud-hosted AI agents.



Become an industry-ready Gen AI Developer from the ground up.



Master the end-to-end lifecycle: design, build, and deploy advanced solutions.



Advance from Python and Deep Learning to context-aware apps with LangChain and RAG.



Customize local LLMs and build autonomous workflows with Crew AI and n8n.



Evaluate and deploy your AI systems to the cloud using Docker and FastAPI.



Foundational Skills

Build a job-ready foundation in Python, Object-Oriented Programming (OOP), and Data Science.



Deep Learning Mastery

Master core architectures including CNNs, RNNs, Transformers, and Feedforward networks with Backpropagation.



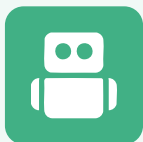
Practical AI Workflows

Design Gen AI applications for NLP and Computer Vision through hands-on projects with Hugging Face and LangChain.



AI System Design

Work with Embeddings, Vector Databases, and complex RAG pipelines to leverage your own data.



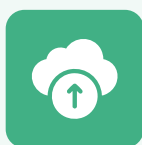
LLM Customization & Agents

Customize Large Language Models using QLoRA and PEFT, and build sophisticated multi-agent systems.



Responsible AI

Build secure and reliable solutions by implementing guardrails, bias checks, and rigorous evaluation metrics.



End-to-End Deployment

Design, build, and deploy full AI solutions using FastAPI, Docker, local LLMs, and Streamlit/Gradio interfaces.



Skills You'll Gain



Prompt Engineering



Deep Learning & Transformers



RAG Systems



LLM Fine-tuning



AI Agents



Model Context Protocols (MCP)



Model Evaluation



MLOps & Deployment

AI Tools You Will Learn



Hugging Face



LangChain



Ollama



Crew AI



n8n



FastAPI



Gradio & Chainlit



Docker



DeepEval & LangSmith

Learn From Industry Experts

Guided by leaders actively building and deploying enterprise AI solutions.



Sampurn Rattan Jain

Data Science - Solution Architect



Experienced Architect: Over 10 years of experience designing end-to-end Data Science and AI solutions, with a strong focus on the BFSI landscape.



Domain Expertise: Diverse background delivering AI solutions across Energy Management, Healthcare, EdTech, and national security.



Advanced AI Implementations: Recently spearheaded the creation of virtual workers using multi-agentic frameworks for a large insurance organization.



Current Leadership: Leads the Data Science Practice at GeeksForGeeks, optimizing internal processes and mentoring professionals.



Ashish Jangra

Data Science Mentor



Experienced Educator: Over 5 years of experience mentoring and educating in the fields of Data Science and AI.



Massive Reach: Has successfully trained over 20,000 students across technologies like Machine Learning, Deep Learning, and NLP.



Innovative Content Creator: Authored India's first Self-Driving Cars course alongside Elite Techno Groups.



Current Role: Actively serves as a Data Science Mentor at GeeksforGeeks, with a background as an AI trainer for Teenage Coder.

16-Week

Curriculum Journey

Structured for working professionals with weekend/evening live sessions.

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Phase 1: Weeks 1-3

Foundational Knowledge

Establish a rock-solid foundation in Python and deep learning architectures before moving into generative models.

- Python Basics for Data Science
- Feedforward Neural Networks & Backpropagation
- Convolutional Neural Networks (CNNs)
- Intuition of Recurrent Neural Networks (RNNs)

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Phase 2: Weeks 4-7

Core Generative AI Concepts

Master the models that power modern AI and the frameworks used to control them.

- Transformers Conceptual Knowledge (Attention, Encoders, Decoders)
- Hugging Face ecosystem & Pipeline Abstraction
- Introduction to LangChain & LCEL
- Transfer Learning & Fine-tuning (QLoRA, PEFT)
- Prompt Engineering Techniques

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Phase 3: Weeks 8-13

Advanced Architectures & Agents

Build systems that can search private data, run locally, and autonomously execute tasks.

- Embeddings & Vector Databases
- Retrieval Augmented Generation (RAG, HyDE, Hybrid)
- Running Local Large Language Models (Ollama)
- Agentic AI with Crew AI (Multi-agent Systems)
- Model Context Protocol (MCP) & Agent Skills
- Low-Code Agentic AI using n8n

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Phase 4: Weeks 14-16 + Bonus

Production, Evaluation & Deployment

Turn experimental notebooks into reliable, containerized, cloud-hosted applications.

- LLM Evaluation & Observability (DeepEval, LangSmith)
- FAST APIs and Pydantic Schemas
- UI for LLMs (Gradio, Chainlit)
- Containerization with Docker
- Deployment (Vercel, Cloud Run, HF Spaces)
- Bonus Week: Capstone Review with Industry Pros



Build a World-Class Portfolio

Stop doing toy problems. Build the applications companies are hiring for today.



Text-to-SQL App

Create a system that allows users to query databases using natural language, featuring self-querying RAG and schema validation.



Autonomous Market Research Agent

Orchestrate a multi-agent system using Crew AI that autonomously scrapes data, analyzes trends, and writes reports.



AI Newsletter (n8n)

Build a low-code automated workflow using n8n that aggregates news, summarizes content via LLMs, and formats a newsletter.



Semantic Search Engine

Build a powerful semantic search engine utilizing Vector Databases and advanced embedding techniques on unstructured data.



Multimodal Text-Image App

Develop an intelligence application that processes, analyzes, and reasons across both text and visual data simultaneously.



QLoRA Fine-Tuned LLM

Apply Parameter-Efficient Fine-Tuning (PEFT) and QLoRA techniques to fine-tune an open-source LLM on specific domain data.



Responsible AI Framework

Implement guardrails, conduct bias checks, and deploy rigorous evaluation metrics utilizing DeepEval and LangSmith.



End-to-End Capstone

Design, build, evaluate, and deploy a complete Generative AI system packaged in Docker and deployed to a modern cloud platform.



Unmatched Support System

We ensure you never get stuck. Our ecosystem is designed to support you from day one to deployment.



Live Mentor Interaction

Direct access to industry professionals during live sessions.



24x7 AI Doubt Support

Instant help at 2 AM from our custom trained AI teaching assistant.



Limited Batch Size & Weekly Q&A

Online doubt classes every Tuesday & Thursday in intimate settings.

Industry Recognized





From Classes to Careers

Mastering these workflows positions you for the most in-demand roles in modern software development.

Generative AI Engineer

AI Automation Developer

LLM Application Developer

AI Solutions Engineer

Where Our Alumni Work



Microsoft



Google



Amazon



Accenture

Infosys

Infosys



Intel



Capgemini



Cognizant



Wipro



Salesforce



Airtel



TCS



Cisco



HCL



Oracle



Flipkart



Paytm



HP



Dell



Goldman
Sachs

 Includes Resume/LinkedIn Optimization and Mock Interviews

What Our Students Say

“This course is a game-changer! The skills I acquired here have already opened up new career opportunities for me. The mentors go above and beyond to prepare you for real-world scenarios!”

VK Vatsal Kumar

“I can't express how much I've gained from this course. The mentors are top-notch, and the course content is comprehensive. They break down complex concepts into easy-to-understand modules...”

SG Shamita Gupta

“I can't believe how much I've learned. It covers everything from basics to advanced analytics, and the progression is smooth. The practical exercises and projects are what sets this course apart.”

S Sarah

“Nice and informative! Clear explanations and helpful guidance throughout!”

AK Aniket Kokate



Frequently Asked Questions

? Do I need prior experience?

Basic programming knowledge in Python is recommended. The course starts with deep learning foundations before moving to advanced Gen AI topics.

? Will I learn how to deploy applications?

Yes! You will deploy AI APIs using FastAPI, Docker, local LLMs, and build interfaces with Streamlit and Gradio.

? Does the program cover Agentic AI?

Yes. You will design autonomous AI systems using frameworks like CrewAI and orchestrate low-code workflows with n8n.

? What is the final outcome?

You will design, build, evaluate, and deploy a complete end-to-end Generative AI system as your capstone showcase, ready for your portfolio.

What is the final outcome?

Join the live program and accelerate your career in Generative AI.

Call Us

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Geeks Learning Together!

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